

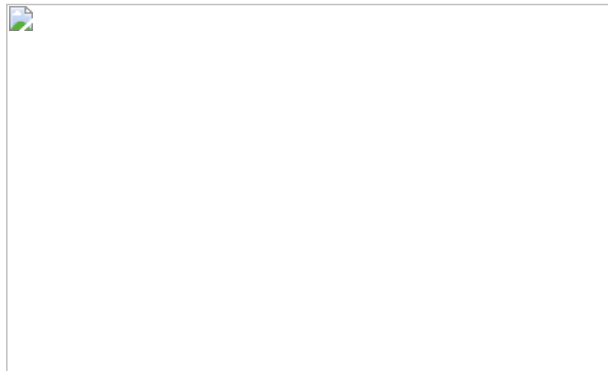
Crop Production :: Pulses :: Blackgram

BLACKGRAM (*Vigna mungo* L.)



1. SEASON AND VARIETIES

District/Season	Varieties
Adipattam (June-August) All districts except Kanyakumari and Nilgiris	VBN (Bg) 4, VBN(Bg) 5, TNAU (Black gram) VBN 7
Puratasipattam (September-November) Vellore, Tiruvannamalai Dharmapuri, Salem, Namakkal, Perambalur, Erode, Coimbatore, Madurai, Dindigul, Theni, Pudukottai., Sivagangai, Ramanathapuram, Virudhunagar, Thoothukudi and Tirunelveli.	VBN 3 (To avoid Yellow Mosaic Virus treat the seeds with Imidachlorphid @ 1 ml / kg seed) VBN(Bg) 4, VBN (Bg) 5, TNAU (Blackgram) VBN 6, TNAU (Blackgram) Co 6, APK 1
Markazhi – Thaipattam(Winter Irrigated) All districts except Kanyakumari and Nilgiris	VBN 3, VBN (Bg) 4, VBN (Bg) 5, TNAU (Blackgram) VBN 6, TNAU (Blackgram) Co 6, TMV 1
Rice fallows (January) Thanjavur, Tiruvarur, Nagapattinam, Cuddalore, Villupuram and Kanchipuram	ADT 3
Chithiraipattam (Summer Irrigated) Thanjavur, Tiruvarur, Nagapattinam, Cuddalore, Villupuram, Tiruchirappalli, Perambalur, Thiruvallur, Kancheepuram	ADT 5



TNAU (Blackgram) Co 6

MANAGEMENT OF FIELD OPERATIONS

DESCRIPTION OF BLACKGRAM VARIETIES

Particulars	T 9	CO 5	ADT 3	VBN 1
Parentage	Selection from Bareilly, U.P	Pureline selection from Musiri -	Pureline selection from Tirunelveli local	KM 1xH 76-1
Year of release	1972	1981	1981	1987
50% flowering (days)	30 - 35	35 - 40	30 - 35	30 - 35
Maturity duration (days)	65 - 70	70 - 75	70 - 75	60 - 65
Grain yield (kg/ha)				
Rainfed	-	740	..	700
Irrigated	1000	1270	-	850
Rice fallows			720	
Height (cm)	35 - 40	30 - 35	50	30 - 35
Clusters	10 - 12	10 to 12	10 - 15	12 - 13
Hairiness of pods	Glabrous	Hairy	Hairy	Hairy
Colour of grain	Black & Dull	Black & Dull	Black & dull	Black
100 grain wt (g)	4.0	5.7	3.6	5.1

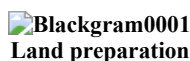
Particulars	ADT 5	K 1	VBN 2	VBN 3	VBN(Bg) 4
Parentage	Selection from Kanpur variety	CO 3 x US 131	Spontaneous mutant	LBG 402 x LBG 17	CO 4 x PDU 102
Year of release	1988	1994	1996	2000	2003
50% flowering (days)	32	40 - 45	33 - 40	35 - 40	35 - 40
Maturity duration (days)	62	70 - 75	60 - 70	65 - 70	75 - 80
Grain yield (kg/ha)	-	707	750	775	780
Rainfed					
Irrigated	1323	-	1000	825	900
Rice fallows		-	-		
Height (cm)	20 - 25	30 - 35	25 - 35	25 - 35	40 - 45
Clusters	13 - 15	10 - 13	4 - 18	5 - 10	8 - 10
Hairiness of pods	Hairy	Hairy	Glabrous	Hairy pods	Glabrous

Colour of grain	Black	Dull black	Black with green tinge	Dull black	Black
100 grain wt (g)	3.6	5.3	3.83	4.4	4.8

Particulars	VBN(Bg) 5	TNAU (Blackgram) VBN 6	TNAU (Blackgram) VBN 7
Parentage	Vamban 1 x UK 17	Vamban 1 x <i>Vigna mungosilvestris</i>	Vamban 3 x <i>Vigna mungosilvestris</i>
Year of release	2007	2011	2012
Maturity duration (days)	65-70	65-70	65-70
Grain yield (kg/ha)	836	850	882
Rainfed			
Irrigated	820	890	981
Height (cm)	34	18.6	20
Hairiness of pods	Hairy	Hairy	Hairy
100 grain wt (g)	4.0	3.8-4.0	3.8 – 4.0
Special features	Moderately resistant to Yellow Mosaic	Resistant to Yellow Mosaic, synchronized pod maturity	Resistant to Yellow Mosaic, synchronized pod maturity

1. FIELD PREPARATION

- Prepare the land to fine tilth and form beds and channels.
- Amendments for soil surface crusting: To tide over the soil surface crusting apply lime at the rate of 2t /ha along with FYM at 12.5 t/ha or composted coirpith at 12.5 t/ha to get an additional yield of about 15 - 20%.



SEED RATE

Quantity of seed required kg/ha

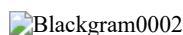
STRAIN	Pure crop	Mixed crop
T 9, CO 5, TMV 1, VBN 1, VBN 2, VBN 3, VBN (Bg) 4 ADT 5, TMV 1	20	10
(Rice fallows) ADT 3	25	..

Optimum plant population 3,25,000/ha

2. SEED TREATMENT

Treat the seeds with Carbendazim or Thiram @ 2 g/kg of seed 24 hours before sowing (or) with talc formulation of *Trichoderma viride* @ 4g/kg of seed (or) *Pseudomonas fluorescens* @ 10 g/kg seed. Bio control agents are compatible with biofertilizers. First treat the seeds with Biocontrol agents and then with Rhizobium. Fungicides and biocontrol agents are incompatible.

Note: Seed treatment will protect the seedlings from seed borne pathogens, root-rot and seedlings diseases.



Source: http://upload.wikimedia.org/wikipedia/commons/6/6f/Black_gram.jpg

3. SEED TREATMENT WITH BIOFERTILIZER

Treat the seeds with 3 packets (600 g/ha) of Rhizobial culture CRU-7 + 3 packets (600 g/ha) of PGPR and 3 packets (600 g/ha) of Phosphobacteria developed at TNAU using rice kanji as binder. If the seed treatment is not carried out apply 10packets of Rhizobium (2000 g/ha) + 10 packets of PGPR (2000 g/ha) and 10 packets (2000 g) of Phosphobacteria with 25 kg of FYM and 25 kg of soil before sowing.

4. FERTILIZER APPLICATION

- Apply fertilizers basally before sowing.
Rainfed : 12.5 kg N + 25 kg P₂O₅ + 12.5 kg K₂O +10 kg S*/ha
Irrigated : 25 kg N + 50 kg P₂O₅ + 25 kg K₂O + 20 kg S*/ha

*Note : Applied in the form of gypsum if Single Super Phosphate is not applied as a source of phosphorus

- Soil application of 25 kg ZnSo₄/ha under irrigated condition
- Soil application of TNAU micronutrient mixture @ 5 kg/ha as Enriched FYM (Prepare enriched FYM at 1:10 ratio of MN mixture & FYM ; mix at friable moisture & incubate for one month in shade).

Foliar spray of 1% urea for yield improvement in black gram

For yield improvement through increasing the physiological, biochemical attributes, foliar spray of urea 1% on 30 and 45 days after sowing is recommended. For rice fallow pulses in Delta area, the present recommendation of foliar spray of 2% DAP may be continued.

Foliar spraying to mitigate moisture stress

Foliar spraying of 2% KCl + 100 ppm Boron during dry spell as mid season management practice in black gram during *Rabi* season is recommended to increase the yield over KCl spray alone .

Economizing the use of micronutrients through seed treatment for blackgram

Seed coating with biofertilizers and micronutrients viz., Zn, Mo & Co @ 4, 1, 0.5 g/kg of seed is recommended.

Nitrogen substitution by organic sources for pulses

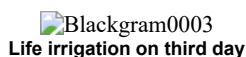
50 per cent nitrogen can be substituted through organic source (850 kg of vermicompost per hectare). Lime application is recommended for pulses with soil pH less than 6.0.

5. SOWING OF SEEDS

1. For irrigated crop dibble the seeds adopting 30 x 10 cm spacing
2. For rainfed crop dibble the seeds adopting 25 cm x 10 cm spacing

6. WATER MANAGEMENT

Irrigate immediately after sowing, followed by life irrigation on the third day. Irrigate at intervals of 7 to 10 days depending upon soil and climatic conditions. Flowering and pod formation stages are critical periods when irrigation is a must. Avoid water stagnation at all stages. Apply KCl at 0.5 per cent as foliar spray during vegetative stage if there is moisture stress.



7. SPRAYING OF DIAMMONIUM PHOSPHATE OR UREA, NAA AND SALICYLIC ACID

- a. Foliar spray of NAA 40 mg/lit and Salicylic acid 100 mg/lit once at pre-flowering and another at 15 days thereafter
- b. i) For rice fallow crops foliar spray of DAP 20 g/lit once at flowering and another at 15 days thereafter
ii) For irrigated and rainfed crops, foliar spray of DAP 20 g/litre or Urea 20 g/litre once at flowering and another at 15 days thereafter.
- c. Foliar spray of salicylic acid 100 mg/litre once at preflowering and another at 15 days there after.

8. WEED MANAGEMENT

1. Pre emergence application of Pendimethalin 3.3 litres/ha under irrigated condition 2.5 litres/ha under rainfed condition on 3 days after sowing using Backpack/ Knapsack/Rocker sprayer fitted with flat fan nozzle using 500 litres of water for spraying one ha followed by one hand weeding at 20 DAS (or) EPOE application of quizalofop ethyl @ 50 g ai/ha-1 and imazethapyr @ 50 g ai ha-1 on 15 – 20 DAS. If herbicides are not applied give two hand weedings on 15 and 30 days after sowing.
2. For the irrigated blackgram PE isoproturon @ 0.5 kg ha-1 followed by one hand weeding on 30 DAS.

9. Multi bloom technology

A special technology being practiced in Pattukottai block of Tanjore district for blackgram and greengram. The soil is alluvial and rich in organic matter and nutrients. The crop is sown during early summer (Jan.-Feb.) as normal crop and fertilizer is applied as per the recommendation for irrigated crop. In addition to that, top dressing of Nitrogen is done with an extra dose of 25 to 30 kg through urea. Since pulses are indeterminate growth habit and continue to produce new flashes the top dressing will be done on 40-45 days after sowing. The crop complete its first flesh of matured pods during 60-65th day, further their second new flesh within 20-25 days. Therefore two fleshes of pods can be harvested at a time within the duration of 100 days.

RICE-FALLOWS

VARIETIES AND SEED RATE

Quantity of seed required kg/ha		
Varieties	Sole crop	Mixed crop
CO 4, ADT 2, ADT 3, ADT 4, ADT 5, TMV 1 (Rice fallows)	30	..

1.TIME OF SOWING

Third week of January –Second week of February

2.SOWING OF SEEDS

1. For relay cropping broadcast the seeds in the standing crop 5 to 10 days before the harvest of the paddy crop uniformly under optimum soil moisture conditions so that the seeds should get embedded in the waxy mire.
2. For combined harvesting areas, broadcast the seeds before harvesting the paddy crop with machinery

3. SPRAYING OF DIAMMONIUM PHOSPHATE , NAA AND SALICYLIC ACID

- Foliar Spray of NAA 40 mg/lit and Salicylic acid 100 mg/lit once at pre-flowering and another at 15 days thereafter
- Foliar spray of DAP 20 g/lit once at flowering and another at 15 days thereafter
- Foliar spray of salicylic acid 100 mg/litre once at preflowering in another and 15 days there after.

4. HARVESTING

- Picking the matured pods, drying and processing
- Uprooting or cutting the whole plants ,heaping ,drying and processing

Source:

[www.upload.wikimedia.org/wikipedia/commons/6/6f/Black_gram.jpg](https://upload.wikimedia.org/wikipedia/commons/6/6f/Black_gram.jpg)

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